



Oil, Earth mass and gravitational force

Khaled Moustafa

Conservatoire National des Arts et Métiers, Paris, France



HIGHLIGHTS

- Large amounts of fossil fuels are extracted annually worldwide.
- Would the extracted amounts represent any significant percentage of the Earth mass?
- What would be the consequence on Earth structure and its gravitational force?
- Modeling the potential loss of Earth mass might be required.
- The efforts for alternative renewable energy sources should be enhanced.

ARTICLE INFO

Article history:

Received 21 December 2015

Received in revised form 20 January 2016

Accepted 21 January 2016

Available online 3 February 2016

Editor: D. Barcelo

Keywords:

Earth mass

Fossil fuels

Oil depletion

Gravitational force

Consequence oil depletion

Oil extraction

Climate change

ABSTRACT

Fossil fuels are intensively extracted from around the world faster than they are renewed. Regardless of direct and indirect effects of such extractions on climate change and biosphere, another issue relating to Earth's internal structure and Earth mass should receive at least some interest. According to the Energy Information Administration (EIA), about 34 billion barrels of oil (~4.7 trillion metric tons) and 9 billion tons of coal have been extracted in 2014 worldwide. Converting the amounts of oil and coal extracted over the last 3 decades and their respective reserves, intended to be extracted in the future, into mass values suggests that about 355 trillion tons, or $\sim 5.86 \times 10^{-9}$ (~0.0000000058)% of the Earth mass, would be 'lost'. Although this is a tiny percentage, modeling the potential loss of Earth mass may help figuring out a critical threshold of mass loss that should not be exceeded. Here, I briefly discuss whether such loss would have any potential consequences on the Earth's internal structure and on its gravitational force based on the Newton's law of gravitation that links the attraction force between planets to their respective masses and the distance that separate them.

© 2016 Elsevier B.V. All rights reserved.

Million years ago, dense populations of plants, animals and microorganisms have lived, perished and sedimented as organic matter. High temperatures and pressures 'cooked' the accumulating organic matter over a long time and produced fossil fuels in the form of coal, oil, and gases embodied in sedimentary rocks as integral parts of rock structures and Earth mass. The first commercial oil wells have been drilled in 1850s (Taverne, 2008). Since then, fossil fuels have been extensively extracted in tremendous amounts from around the world to fulfill the energy needs of the Industrial Revolution. According to the US Energy Information Administration (EIA) (<http://www.eia.gov>), the production of oil in 2014 was estimated at about ~93 million barrels/day (Table 1). Considering that one barrel of oil weighs about ~140 kg in average (depending on oil grades and density), the weight of oil pumped out in 2014 would be: $93,000,000 \times 140 \times 365 = 4,752,300,000,000$ kg (~4.75 trillion tons). During the last 34 years (from 1980 to 2014), the

total amount of oil produced worldwide is estimated at about ~132 trillion tons (Table 1).

The production of coal, for its part, has significantly increased from 1980 to 2012. In 1980, it was about 4 billion tons, and in 2012 it has almost doubled (~9 billion tons) (Table 2). The total production of coal during the last 32 years (1980–2012) is estimated at about ~186 billion (0.186 trillion) tons (Table 2).

The total mass of oil and coal extracted during the last 34 years (1980–2014) would thus be: 132 trillion ton oil + 0.186 trillion ton coal = 132.186 trillion tons (~132,186,000,000,000 kg).

On the other hand, the reserve of oil is estimated at about 1,655,560,000,000 barrels¹ (~231,779,000,000,000 kg) and the reserve of coal at about 979,791,000,000 kg.² The total reserves of both oil and

E-mail address: khaled.moustafa@gmail.com.

¹ <https://www.eia.gov/cfapps/ipdbproject/iedindex3.cfm?tid=5&pid=57&aid=6&cid=ww&syid=2014&eyid=2015&unit=BB>. Accessed 20 January 2016.

² <https://www.eia.gov/cfapps/ipdbproject/iedindex3.cfm?tid=1&pid=7&aid=6&cid=ww&syid=2011&eyid=2011&unit=MST>. Accessed 20 January 2016.

Table 1
Total Oil Production worldwide from 1980 to 2014.^a

Year	Oil barrel per day	Oil barrel per year (365 days)	Oil weight (kg) per barrel (1 barrel = ~140 kg)
2014	93,201,088	34,018,397,029	4.76258E + 12
2013	91,014,472	32,859,632,846	4.27175E + 12
2012	90,466,487	32,737,529,404	4.25588E + 12
2011	88,532,260	32,059,502,305	4.16774E + 12
2010	88,099,485	31,946,591,792	4.15306E + 12
2009	85,703,404	31,006,494,741	4.03084E + 12
2008	86,514,617	31,287,562,535	4.06738E + 12
2007	85,130,475	30,862,881,629	4.01217E + 12
2006	85,134,828	30,890,356,212	4.01575E + 12
2005	85,099,148	30,896,112,587	4.01649E + 12
2004	83,402,075	30,353,438,358	3.94595E + 12
2003	79,606,389	29,056,332,127	3.77732E + 12
2002	77,100,672	28,141,745,138	3.65843E + 12
2001	77,672,247	28,350,370,199	3.68555E + 12
2000	77,725,453	28,369,790,498	3.68807E + 12
1999	74,838,478	27,316,044,346	3.55109E + 12
1998	75,680,734	27,623,467,855	3.59105E + 12
1997	74,219,760	27,090,212,579	3.52173E + 12
1996	71,986,085	26,274,920,890	3.41574E + 12
1995	70,304,631	25,661,190,425	3.33595E + 12
1994	68,636,649	25,052,376,739	3.25681E + 12
1993	67,101,335	24,491,987,260	3.18396E + 12
1992	66,552,553	24,291,682,002	3.15792E + 12
1991	66,339,100	24,213,771,606	3.14779E + 12
1990	66,435,706	24,249,032,577	3.15237E + 12
1989	65,518,811	23,914,365,960	3.10887E + 12
1988	64,394,518	23,503,998,975	3.05552E + 12
1987	62,099,487	22,666,312,587	2.94662E + 12
1986	61,533,681	22,459,793,718	2.91977E + 12
1985	59,156,363	21,592,072,594	2.80697E + 12
1984	59,563,225	21,740,577,209	2.82628E + 12
1983	57,927,660	21,143,595,915	2.74867E + 12
1982	58,097,931	21,205,744,676	2.75675E + 12
1981	60,602,099	22,119,766,036	2.87557E + 12
1980	63,987,116	23,355,297,504	3.03619E + 12
Total (kg)			1.32317E + 14
Total oil weight in trillion tons			132 trillion tons

^a Source: Energy Information Administration (USA): <http://www.eia.gov/cfapps/ipdbproject/iedindex3.cfm?tid=5&pid=53&aid=1&cid=ww.&syid=1980&eyid=2015&unit=TBPd> Accessed on 20 January 2016.

coal would thus be: 231,779,000,000,000 kg Oil + 979,791,000,000 kg Coal = 232,758,790,000,000 kg (~233 trillion tons). Subsequently, the total sum of oil and coal extracted during the last 34 years and their respective reserves would be $132 + 233 = 355$ trillion tons (or 350,000,000,000,000 kg) ($\sim 3.5 \times 10^{14}$ kg).

By reporting this value to the value of the Earth's mass estimated at about 5.97×10^{24} kg³ (5,972,190,000,000,000,000,000 kg), which includes the mass of the atmosphere estimated at about 5.1480×10^{18} kg (Trenberth and Smith, 2005), we find that the mass of the oil and coal already extracted during the last 34 years and their respective reserves, intended to be extracted in the future, would represent about 5.86×10^{-9} (~0.0000000058) % of the global Earth mass. In other words, about 0.0000000058% of the Earth mass would be 'lost' in the form of burned oil and coal. Although this percentage may appear tiny, a small percentage of a gigantic number such as the Earth's mass would represent an important overall value. Moreover, the real amounts of the extracted oil and coal should be greater than the estimated here, if we should take into account the amounts of oil and coal that have been extracted before 1980, for which no relevant statistics are available.

As such, the questions that could be raised here are: what would be the potential consequences of fossil fuel extraction and burning on Earth's mass at the long-term? What would be the effects on the Earth's internal structure and on its gravitational force in regard with

Table 2
Total coal production worldwide from 1980 to 2012*.

Year	Coal production (tons)
2012	8,687,297,074
2011	8,443,802,731
2010	7,999,454,814
2009	7,601,609,006
2008	7,470,959,169
2007	7,235,882,783
2006	6,965,038,273
2005	6,636,340,567
2004	6,216,305,801
2003	5,813,253,141
2002	5,429,459,293
2001	5,375,192,780
2000	5,137,687,939
1999	4,978,418,246
1998	5,039,867,508
1997	5,069,638,967
1996	5,047,504,716
1995	4,955,331,148
1994	4,872,044,207
1993	4,794,646,914
1992	4,929,168,603
1991	5,018,521,216
1990	5,345,000,328
1989	5,311,262,411
1988	5,234,804,720
1987	5,119,885,558
1986	5,009,016,691
1985	4,891,863,416
1984	4,665,243,152
1983	4,415,548,925
1982	4,379,196,923
1981	4,221,808,991
1980	4,179,632,567
Total (tons)	1.86491E + 11
(Total in trillion tons)	(~0.186 trillion tons)

* Source: Energy Information Administration (USA): <http://www.eia.gov/cfapps/ipdbproject/iedindex3.cfm?tid=1&pid=7&aid=1&cid=ww.&syid=1980&eyid=2012&unit=TST> Accessed on 20 January 2016.

neighboring planets and stars based on Newton's universal law of gravitation that links the gravitational force proportionally to the mass? According to Newton's law, two planets in the universe attract each other with a gravitational force (F) that is proportional to their corresponding masses but inversely proportional to the square of the distance between them (Fig. 1). The force (F) can be calculated through the equation: $F = \frac{Gm_1m_2}{r^2}$, where 'F' is the attraction force between two masses (m_1 and m_2), 'G' is a gravitational constant and 'r' is the distance between the two planets or masses (from center-to-center). Subsequently, if the Earth would lose some of its mass in the form of oil and coal burned overtime, the question is: would Earth gravitational force and its rotation be impacted in whatever way with respect to other planets or stars? If so, how, and to which extent such impacts could be avoided or reverberated on Earth?

One, however, may argue that the potential loss of Earth mass in the form of burned fossil fuels would fit in Lavoisier's Law, "*Nothing is lost, ... everything is transformed*", but extracting enormous amounts of fossil fuels from Earth would move a large part of Earth mass from one part (Earth interior) to another (Earth surface and atmosphere in the form of gases; mainly CO₂), potentially resulting in the fragilization of the Earth's internal structure relative to its state upon formation about 4.5 billion years ago (Dalrymple, 2001). Additionally, the major part of the extracted fossil fuels is burned, and therefore, one metric ton of extracted fossil fuels will never produce one metric ton of counterpart products to eventually maintain comparable or proportional masses between the extracted and transformed fossil fuels. In other words, the loss of Earth's mass by burning oil and coal would not be compensated by new deposits of oil and coal as quickly as they are depleted nor by similar mass of transformed byproducts obtained by industrial

³ <http://solarsystem.nasa.gov/planets/earth/facts>. Accessed 20 January 2016.

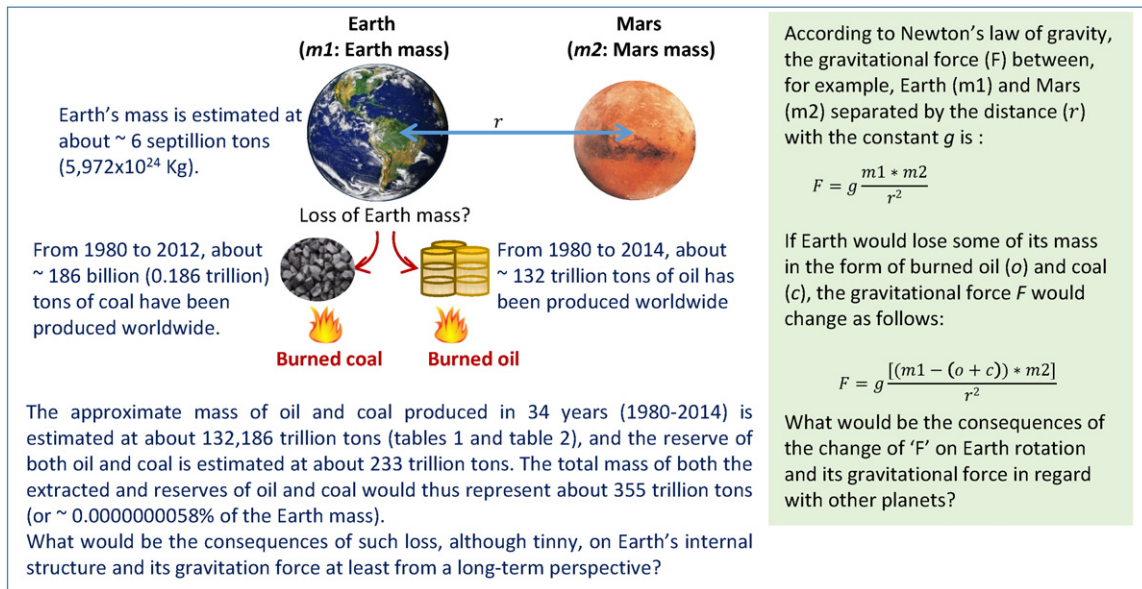


Fig. 1. According to Newton's universal gravity law, planets attract each other through a force (F) that is proportional to their masses and inversely proportional to the distance between them (r). By extracting and burning trillions tons of oil and coal during centuries, Earth would lose some percentage of its mass. What would be the consequence of such losses on Earth's gravitational force vis-à-vis other planets, particularly neighboring ones (i.e. Venus and Mars) or between Earth and moon or Sun? If such a loss would have any potential effect on Earth gravitational force, then what would prevent neighboring planets from 'failing' on Earth, or inversely, if Earth is losing some of its mass progressively, where the mass is a determinant factor in maintaining gravitational relationship with other planets based on Newton's law of gravitation?

operations to ultimately maintain the same global mass or the same terrestrial attraction force related to its final mass. Extensive extractions of oil and coal over large regions may also 'fragilize' their internal or underground structure and would increase the rate and intensity of Earthquakes as the huge rocky spaces filled by oil and coal are fully exhausted and remain either empty or filled with other liquids or water whose density is different from oil and coal, hence exerting different pore pressures that may affect the earthquake frequency and magnitude. Recent studies tend to support such claims particularly nearby the water-injected sites (Ellsworth, 2013; Frohlich, 2012; Keranen et al., 2013; Keranen et al., 2014; van der Elst et al., 2013). It was also reported that earthquakes could be induced either by fluid injection or by fluid extraction (Segall, 1992; Nicholson and Wesson, 1992; Yerkes and Castle, 1976). In fact, seismicity would be correlated in time and space with oil production from some oil and gas fields where pore pressures may decline substantially by several tens of megapascals (Segall,

1989) inducing seismic instabilities at both local and regional scales (Grasso, 1992). On the other hand, the interest in developing industry of unconventional natural gases such as the shale gas continues to grow with potential adverse impacts on the environment and public health (Hays et al., 2015; Meng, 2015; Prpich et al., 2015) although this is still debatable (Werner et al., 2015).

It took billions years to form energy fossils as integral parts of Earth's structure and consistency. However, as the internal activity of Earth evolves permanently (Davies, 1999) and as human activities significantly and rapidly modify the rate and timing of geomorphic processes (Goff et al., 2015), the chemical and geophysical changes caused by oil extraction and depletion may affect Earth dynamism and its subsequent evolvability. It is very likely that enough time has not been elapsed yet to perceive concrete consequences of the intensive depletion of organic fossils which, however, may trigger one day a series of accelerated and unstoppable geological events, particularly when vast regions are

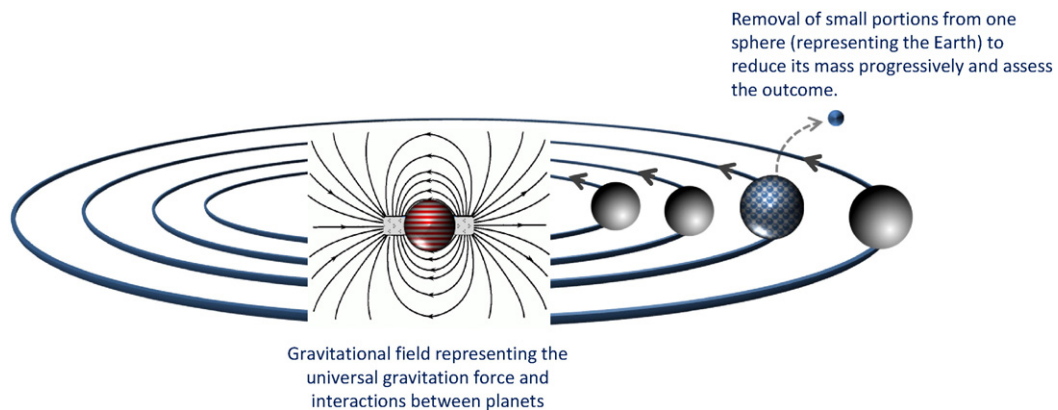


Fig. 2. A streamlined modeling scheme for the potential loss of Earth mass by oil and coal extraction. The solar system could be modeled with magnetic spheres representing planets rotating in virtual orbits under a given current intensity that keeps them in equilibrated distances with each other based on their respective masses and distances. Small portions of the sphere representing Earth could then be removed gradually to mimic the percentage of the extracted fossil fuels. The outcome could be assessed to see if the percentage of burned oil and coal (calculated here as ~0.000000006%) would affect the rotating system. However, the removal of 0.000000006% (~6 μ g) from, for example, a magnetic sphere of 1 kg may not affect its rotation or its attraction force but with a gradual removal of its mass, we could see at which percentage of the mass removal the system starts to be affected to eventually figure out a critical threshold beyond which the extraction of fossil fuels should be reduced or avoided to avoid any potential serious global consequences.

completely depleted. Toward reducing this from happening, other energy sources than fossil fuels may need to be developed urgently. If substantial efforts and research investments comparable to those spent on the development of infrastructure for fossil fuel energy would be devoted to the development of safer and sustainable alternative energy sources (for e.g., solar energy, artificial photosynthesis, etc.), the consequences of oil depletion on the global ecosystem could be delayed, should they cannot be completely avoided.

Although new knowledge for the management of the global ecosystem can be acquired from existing data without needs to new experiments or infrastructure (Specht et al., 2015), a modeling or a simulation of the loss of Earth mass in the form of burned oil and coal (Fig. 2) might be required to assess the importance of the loss of terrestrial mass and to probe its potential effects on Earth structure and on its universal gravitational force in regard to other neighboring planets. Interdisciplinary research and collaborations, spanning geology, physics, astrology and environmental sciences would also be required to gather and analyze existing information and provide insight into the following questions: would the rapid and huge extraction of oil and coal result in a real loss of Earth's mass? If so, how much and what would be the part of mass being lost since the discovery of the first oil wells and coal mines till the complete depletion of their respective reserves in the future? What would happen to the mutual attraction force (F) between Earth and other stars, particularly neighboring ones (for example between Earth and moon or between Earth and Sun. or other planets)? Are there any geological or biological events that would compensate the potential loss of Earth mass by oil extraction and burning? Would for example the growing human population mass estimated at about 287 million tons for adult people (Walpole et al., 2012) play any counterbalancing role to the oil and coal lost parts? One should also wonder what would happen to the Earth's internal structure once the reserves of fossil fuels are completely depleted and what would be the magnitude or consequences on earthquakes when occurring within large regions that have been extensively voided from its 'filling organic matter' (oil).

Finally, the challenge of the intensified uses of fossil fuels is one many other environmental challenges (Davis et al., 2015) that need effective control based on innovative solutions and adaptive management to sustain the non-renewable natural resources and to reduce subsequent risks associated with their depletion at least from the long-term perspective.

Disclosure

The opinions expressed in this article are those of the author only, and they do not necessarily reflect those of the institution.

References

- Dalrymple, G.B., 2001. The age of the Earth in the twentieth century: a problem (mostly) solved. *Geol. Soc. Lond., Spec. Publ.* 190, 205–221.
- Davies, G.F., 1999. *Dynamic Earth: Plates, Plumes and Mantle Convection*. Cambridge University Press.
- Davis, J., O'Grady, A.P., Dale, A., Arthington, A.H., Gell, P.A., Driver, P.D., et al., 2015. When trends intersect: the challenge of protecting freshwater ecosystems under multiple land use and hydrological intensification scenarios. *Sci. Total Environ.* 534, 65–78.
- Ellsworth, W.L., 2013. Injection-induced earthquakes. *Science* 341, 1225942.
- Frohlich, C., 2012. Two-year survey comparing earthquake activity and injection-well locations in the Barnett Shale, Texas. *Proc. Natl. Acad. Sci. U. S. A.* 109, 13934–13938.
- Goff, J., Knight, J., Sugawara, D., Terry, J.P., 2015. Anthropogenic disruption to the seismic driving of beach ridge formation: the Sendai coast, Japan. *Sci. Total Environ.* 544, 18–23.
- Grasso, J.-R., 1992. Mechanics of seismic instabilities induced by the recovery of hydrocarbons. *Pure and Applied Geophysics* 139, 507–534.
- Hays, J., Finkel, M.L., Depledge, M., Law, A., Shonkoff, S.B., 2015. Considerations for the development of shale gas in the United Kingdom. *Sci. Total Environ.* 512–513, 36–42.
- Keranen, K.M., Savage, H.M., Abers, G.A., Cochran, E.S., 2013. Potentially induced earthquakes in Oklahoma, USA: links between wastewater injection and the 2011 Mw 5.7 earthquake sequence. *Geology*.
- Keranen, K.M., Weingarten, M., Abers, G.A., Bekins, B.A., Ge, S., 2014. Induced earthquakes. Sharp increase in Central Oklahoma seismicity since 2008 induced by massive wastewater injection. *Science* 345, 448–451.
- Meng, Q., 2015. Spatial analysis of environment and population at risk of natural gas fracking in the state of Pennsylvania, USA. *Sci. Total Environ.* 515–516, 198–206.
- Nicholson, C., Wesson, R.L., 1992. Triggered earthquakes and deep well activities. *Pure and Applied Geophysics* 139, 561–578.
- Prpich, G., Coulon, F., EJ, A., 2015. Review of the scientific evidence to support environmental risk assessment of shale gas development in the UK. *Sci. Total Environ.* <http://dx.doi.org/10.1016/j.scitotenv.2015.11.026>.
- Segall, P., 1989. Earthquakes triggered by fluid extraction. *Geology* 17, 942–946.
- Segall, P., 1992. Induced stresses due to fluid extraction from axisymmetric reservoirs. *Pure and Applied Geophysics* 139, 535–560.
- Specht, A., Guru, S., Houghton, L., Keniger, L., Driver, P., Ritchie, E.G., et al., 2015. Data management challenges in analysis and synthesis in the ecosystem sciences. *Sci. Total Environ.* 534, 144–158.
- Taverne, B., 2008. *Petroleum, Industry, and Governments: A Study of the Involvement of Industry and Governments in the Production, and Use of Petroleum*. Vol 15. Kluwer Law International.
- Trenberth, K.E., Smith, L., 2005. The mass of the atmosphere: a constraint on global analyses. *J. Clim.* 18, 864–875.
- van der Elst, N.J., Savage, H.M., Keranen, K.M., Abers, G.A., 2013. Enhanced remote earthquake triggering at fluid-injection sites in the midwestern United States. *Science* 341, 164–167.
- Walpole, S.C., Prieto-Merino, D., Edwards, P., Cleland, J., Stevens, G., Roberts, I., 2012. The weight of nations: an estimation of adult human biomass. *BMC Public Health* 12, 439.
- Werner, A.K., Vink, S., Watt, K., Jagals, P., 2015. Environmental health impacts of unconventional natural gas development: a review of the current strength of evidence. *Sci. Total Environ.* 505, 1127–1141.
- Yerkes, R.F., Castle, R.O., 1976. Seismicity and faulting attributable to fluid extraction. *Eng. Geol.* 10, 151–167.